NETWORKS AND SECURITY

Assignment 1

Emmanuel Antwi Nyamaah

Contents

Network Design and Security Implementation Report 2
1. Introduction 2
2. Network Design 2
IP Addressing Scheme LAN 4
WAN 5
Routing Protocol Configuration 5
OSPF Configuration Summary (Figures 3–7) 5
BGP Configuration 7
Testing and Verification 8
Ping and Connectivity Assessments 8
Security Policies 8
Firewall and ACL Implementation 8
ACL Verification 10
Conclusion 12
References 12

Network Design and Security Implementation Report

1. Introduction

As a network administrator at Global Enterprises, a multinational organisation, I was tasked with designing and deploying a secure, scalable network infrastructure using Cisco Packet Tracer. This network connects five critical sites: Headquarters (HQ), Branch Office 1, Branch Office 2, the Data Centre, and a Demilitarised Zone (DMZ). The practical simulation included the development of both physical and logical topologies, configuration of routing protocols, implementation of security policies, and verification of end-to-end connectivity.

This report outlines the entire project lifecycle, from architecture planning to security enforcement. Key topics include:

- LAN and WAN architecture
- IP addressing and subnetting
- OSPF and BGP configuration
- ACL implementation
- Security testing and validation

2. Network Design

All five networks feature a LAN equipped with routers, Layer 2/3 switches, and devices like PCs and servers. Devices are interconnected through suitable Ethernet and serial cabling. The HQ serves as the central hub, directly connected through WAN links to the

other four locations, utilising serial interfaces.

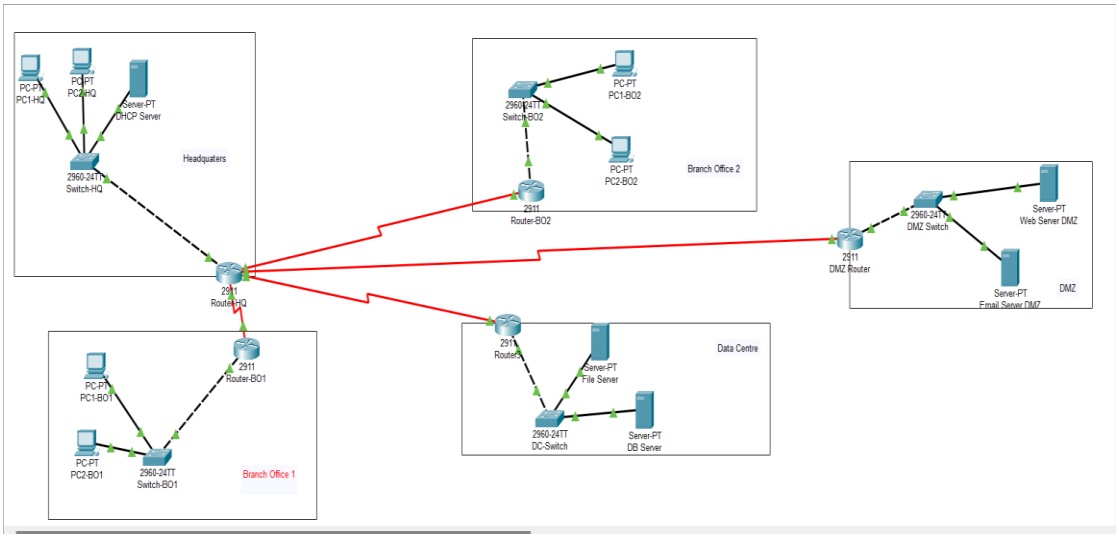


Figure 1- Physical Topology

The 2911 routers and PT switches were selected due to their capability to support dynamic routing protocols and robust security features. Each router was configured with hostnames, interfaces, and clock rates (on DCE serial links), and implemented both static and dynamic routing protocols.

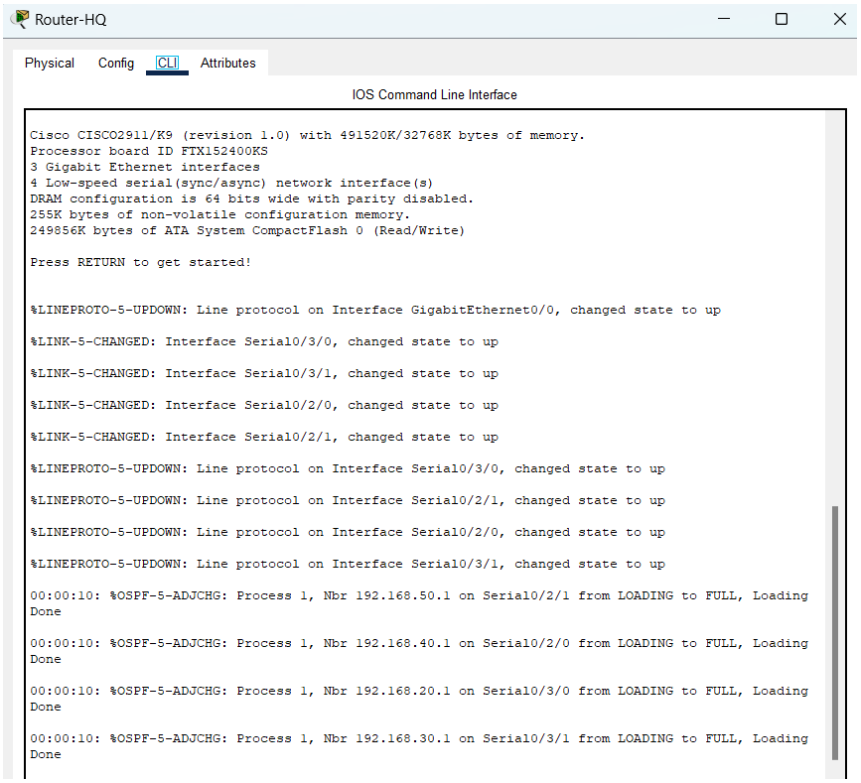


Figure 2 - Headquarters Router configuration

IP Addressing Scheme

LAN

To enhance internal communication in all offices and vital infrastructure zones, I established an organised LAN IP addressing scheme that utilises the private range 192.168.0.0/16, implementing a uniform/24 subnet mask (255.255.255.0) throughout all segments. Each network—Headquarters (HQ), Branch Offices (BO1 and BO2), Data Centre (DC), and the DMZ—received a distinct subnet, with the default gateway configured as the router interface IP (e.g., 192.168.10.1 for HQ).

For instance, the HQ LAN utilises 192.168.10.0/24, with devices like PC1-HQ and the DHCP server assigned IPs such as 192.168.10.2 and 192.168.10.4, all directed to 192.168.10.1 as their gateway. A similar approach is employed across Branch 1 (192.168.20.0/24), Branch 2 (192.168.30.0/24), the Data Center (192.168.40.0/24), and the DMZ (192.168.50.0/24), where each device receives a static IP and is routed through its designated gateway. This consistent addressing method not only provides clarity and separation but also streamlines routing, subnet management, and facilitates future expansion.

Device	Interface	IP address	Subnet mask	Location	Default Gateway
Router-HQ	GigabitEthernet0/0	192.168.10.1	255.255.255.0	HQ LAN	N/A
PC1-HQ	FastEthernet0	192.168.10.2	255.255.255.0	HQ LAN	192.168.10.1
PC2-HQ	FastEthernet0	192.168.10.3	255.255.255.0	HQ LAN	192.168.10.1
DCHP Server	FastEthernet0	192.168.10.4	255.255.255.0	HQ LAN	192.168.10.1
Router-BO1	GigabitEthernet0/0	192.168.20.1	255.255.255.0	BO1 LAN	N/A
PC1-BO1	FastEthernet0	192.168.20.2	255.255.255.0	BO1 LAN	192.168.20.1
PC2-BO1	FastEthernet0	192.168.20.3	255.255.255.0	BO1 LAN	192.168.20.1
Router-BO2	GigabitEthernet0/0	192.168.30.1	255.255.255.0	BO2 LAN	N/A
PC1-BO2	FastEthernet0	192.168.30.2	255.255.255.0	BO2 LAN	192.168.30.1
PC2-BO2	FastEthernet0	192.168.30.3	255.255.255.0	BO2 LAN	192.168.30.1
Router-DC	GigabitEthernet0/0	192.168.40.1	255.255.255.0	DC LAN	N/A
File Server	FastEthernet0	192.168.40.2	255.255.255.0	DC LAN	192.168.40.1
DB Server	FastEthernet0	192.168.40.3	255.255.255.0	DC LAN	192.168.40.1

Router-DMZ	GigabitEthernet0/0	192.168.50.1	255.255.255.0	DMZ LAN	N/A
Web Server	FastEthernet0	192.168.50.2	255.255.255.0	DMZ LAN	192.168.50.1
Email Server	FastEthernet0	192.168.50.3	255.255.255.0	DMZ LAN	192.168.50.1

WAN

For WAN connectivity between the HQ and other network zones, I implemented a dedicated point-to-point addressing scheme utilising the 192.168.x.0/24 private range. Each serial link connecting the routers is assigned a unique subnet with a/24 mask (255.255.255.0), creating isolated routed segments. The router interfaces on both ends have sequential IP addresses for each link to ensure effective route propagation and testing.

For example, the connection between Router-HQ and Router-BO1 employs 192.168.1.1 and 192.168.1.2, whereas the link to BO2 utilises 192.168.2.1 and 192.168.2.2. This point-to-point configuration provides distinct routing paths for dynamic protocols such as OSPF and BGP, enhancing network monitoring and troubleshooting through separated WAN segments.

Link	Device	Interface	IP Address	Subnet Mask
HQ to BO1	Router-HQ	Serial0/3/0	192.168.1.1	255.255.255.0
HQ to BO1	Router-BO1	Serial0/3/0	192.168.1.2	255.255.255.0
HQ to BO2	Router-HQ	Serial0/3/1	192.168.2.1	255.255.255.0
HQ to BO2	Router-BO2	Serial0/3/1	192.168.2.2	255.255.255.0
HQ to DC	Router-HQ	Serial0/2/0	192.168.3.1	255.255.255.0
HQ to DC	Router-DC	Serial0/3/0	192.168.3.2	255.255.255.0
HQ to DMZ	Router-HQ	Serial0/2/1	192.168.4.1	255.255.255.0
HQ to DMZ	Router-DMZ	Serial0/3/0	192.168.4.2	255.255.255.0

Each LAN used DHCP for end devices, while routers and servers used static IPs.

Routing Protocol Configuration

OSPF Configuration Summary (Figures 3–7)

The screenshots together illustrate the complete implementation of OSPF routing (process ID 1) throughout the network's routers: HQ, BO1, BO2, DC, and DMZ. Each router advertises its LAN subnet (e.g., 192.168.10.0, 192.168.20.0, etc.) along with the corresponding WAN point-to-point networks in Area 0, creating a unified OSPF backbone. This configuration allows routers to share routing information dynamically, negating the need for static routes. The %OSPF-5-ADJCHG messages verify the

successful establishment of adjacencies between routers, confirming that OSPF neighbours were identified and that link-state information was appropriately exchanged. This setup supports end-to-end communication across all segments, as demonstrated by successful ping tests, showcasing a well-integrated and scalable, dynamic routing solution ideal for enterprise environments.

```
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.10.0 0.0.0.31 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.2.0 0.0.0.3 area 0
Router(config-router)#network 192.168.3.0 0.0.0.3 area 0
Router(config-router)#network 192.168.4.0 0.0.0.3 area 0
Router(config-router)#
00:37:19: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.20.1 on Se
Done
```

Figure 3 - HQ OSPF Configuration

```
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.20.0 0.0.0.31 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 19
00:36:16: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on Ser
Done
```

Figure 4 - BO1 OSPF Configuration

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.30.0 0.0.0.31 area 0
Router(config-router)#network 192.168.2.0 0.0.0.3 area 0
Router(config-router)#
00:45:02: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on Se
Done
```

Figure 5 - BO2 OSPF Configuration

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.40.0 0.0.0.31 area 0
Router(config-router)#network 192.168.3.0 0.0.0.3 area 0
Router(config-router)#
00:45:38: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on Ser
Done
```

Figure 6 -DC OSPF Configuration

```
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.50.0 0.0.0.31 area 0
Router(config-router)#network 192.168.4.0 0.0.0.3 area 0
Router(config-router)#
00:46:05: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on Se
Done
```

Figure 7 - DMZ OSPF Configuration

BGP Configuration

The screenshot demonstrates the successful configuration and verification of BGP peering between the DMZ router (AS 65010) and the HQ router (AS 65001). The command "show IP BGP summary" confirms that a BGP session is established with the HQ router (192.168.4.1), indicated by the "Up" status and the exchange of 0/0 prefixes, showing an active neighbour relationship. Additionally, the DMZ router has advertised the 192.168.50.0/24 network using the BGP network command. This configuration illustrates a working external BGP setup and verifies the DMZ's capability to share routing information with external ASes, enabling scalable and segmented inter-network communication.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 65010
Router(config-router)#neighbor 192.168.4.1 remote-as 65001
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 192.168.4.1 Up

Router(config-router)#network 192.168.50.0 mask 255.255.255.0
Router(config-router)#
```

Figure 8 - BGP Configurations on DMZ Router

```
Router#show ip bgp summary
BGP router identifier 192.168.50.1, local AS number 65010
BGP table version is 2, main routing table version 6
1 network entries using 132 bytes of memory
1 path entries using 52 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 240 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.4.1     4 65001       5       5        2    0    0 00:03:01        4
```

Figure 9 - BGP Results

Testing and Verification

Ping and Connectivity Assessments

A minimum of 10 ping tests were conducted between PCs and servers across all five networks. Each test validated proper routing and ACL functionality. For instance, while HTTP access to the DMZ server was successful, pings were restricted from Branch 1 according to policy.

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC2-HQ	PC2-BO2	ICMP		0.000	N	0	(edit)	
	Successful	DHCP Server	File Server	ICMP		0.000	N	1	(edit)	
	Successful	PC1-BO1	Web Server DMZ	ICMP		0.000	N	2	(edit)	
	Successful	PC2-BO2	File Server	ICMP		0.000	N	3	(edit)	
	Successful	Email Server DMZ	DB Server	ICMP		0.000	N	4	(edit)	
	Successful	PC2-BO1	PC2-BO2	ICMP		0.000	N	5	(edit)	
	Successful	Router-BO1	PC1-BO2	ICMP		0.000	N	6	(edit)	
	Successful	File Server	Web Server DMZ	ICMP		0.000	N	7	(edit)	
	Successful	PC1-BO2	Email Server DMZ	ICMP		0.000	N	8	(edit)	
	Successful	PC2-BO1	DMZ Router	ICMP		0.000	N	9	(edit)	

Figure 10 -10 ping results after osfp and bgp

Security Policies

Firewall and ACL Implementation

To protect network traffic and mitigate cyber threats, I tailored Access Control Lists (ACLs) and firewall settings for each network zone. On the headquarters router, I restricted remote administrative access by blocking Telnet and SSH from all hosts, except those within a designated trusted IP range, applying these limitations to the VTY lines. For Branch 1 (BO1) and Branch 2 (BO2) routers, I permitted only HTTP and HTTPS outbound traffic to the DMZ web server, while restricting unauthorised access to internal subnets such as the Data Centre. The Data Centre router was configured to allow only necessary management access (SSH and Telnet) from HQ, blocking all other connections. Similarly, the DMZ router was set up to allow only specific web services (ports 80 and 443) while denying ICMP and unauthorized attempts at accessing internal areas. These layered, interface-specific ACLs ensured traffic segmentation, lowered vulnerability, and maintained network access policies tailored to the distinct roles of each location—collectively enhancing the overall security framework of the infrastructure.

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 permit tcp host 192.168.10.2 any eq 22
Router(config)#access-list 100 deny tcp any any eq 22
Router(config)#access-list 100 deny tcp any any eq telnet
Router(config)#line vty 0 4
Router(config-line)#access-class 100 in
Router(config-line)#login
% Login disabled on line 388, until 'password' is set
% Login disabled on line 389, until 'password' is set
% Login disabled on line 390, until 'password' is set
% Login disabled on line 391, until 'password' is set
% Login disabled on line 392, until 'password' is set
Router(config-line)#
```

Figure 11 - ACL Configuration on HQ Router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 110 permit tcp any host 192.168.50.10 eq 80
Router(config)#access-list 110 permit tcp any host 192.168.50.10 eq 443
Router(config)#access-list 110 deny ip any 192.168.50.0 0.0.0.31
Router(config)#interface Serial0/3/0
Router(config-if)#ip access-group 110 out
Router(config-if)#
```

Figure 12 - - ACL Configuration on BO1 Router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 120 deny icmp 192.168.30.0 0.0.0.31 192.168.40.0 0.0.0.31
Router(config)#access-list 120 permit ip any any
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip access-group 120 out
Router(config-if)#
```

Figure 13 - - ACL Configuration on BO2 Router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 130 permit tcp 192.168.10.0 0.0.0.31 host 192.168.40.10 eq 22
Router(config)#access-list 130 deny tcp any host 192.168.40.10 eq 22
Router(config)#access-list 130 permit ip any any
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip access-group 130 in
Router(config-if)#
```

Figure 14 - - ACL Configuration on DC Router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 140 permit tcp any host 192.168.50.10 eq 80
Router(config)#access-list 140 permit tcp any host 192.168.50.10 eq 443
Router(config)#access-list 140 deny ip 192.168.20.0 0.0.0.31 192.168.50.0 0.0.0.31
Router(config)#access-list 140 permit ip any any
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip access-group 140 in
Router(config-if)#
```

Figure 15 - - ACL Configuration on DMZ Router

ACL Verification

The security testing displayed in the screenshots confirms that the Access Control Lists (ACLs) and firewall rules set up across the network are functioning correctly. In the first image, an attempt to access the Data Centre router (192.168.40.1) via Telnet from PC2-HQ was specifically denied, indicating that only pre-approved IPS can access administrative services, validating the VTY line ACL at HQ. In the following screenshots, PC1-BO1 tries to ping both the DMZ router (192.168.50.1) and a Data Centre host (192.168.40.2), but both attempts fail due to ACLs blocking ICMP and TCP traffic from BO1 to these vital infrastructure areas. These results directly corroborate the previously outlined configuration where security was enforced on the BO1 and DMZ routers to prevent unauthorised access to sensitive sections. Overall, these tests confirm that the ACLS are not only syntactically accurate but also effectively safeguards the network by enforcing isolation and access restrictions in accordance with the security policy design.

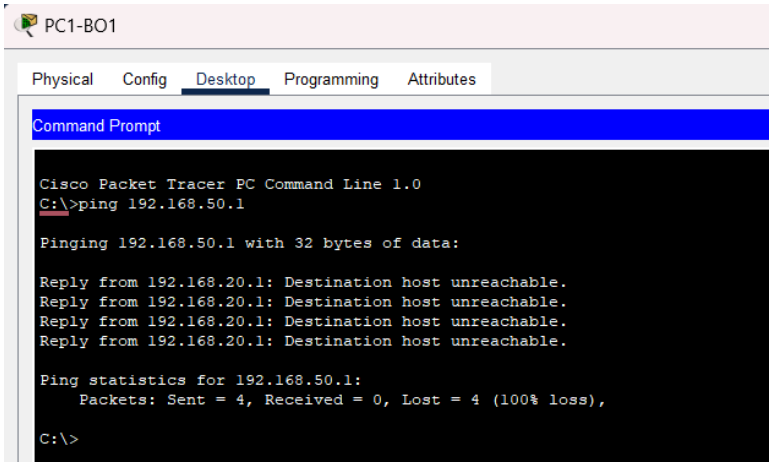


Figure 16 - Pinging Web Server in DMZ from PC2 in Headquarters.

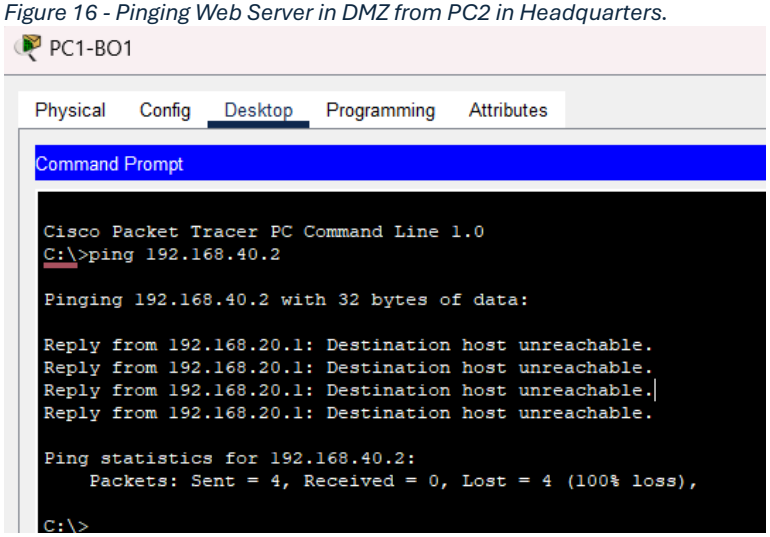


Figure 17 - Pinging File Server in DC from PC1 in BO1.

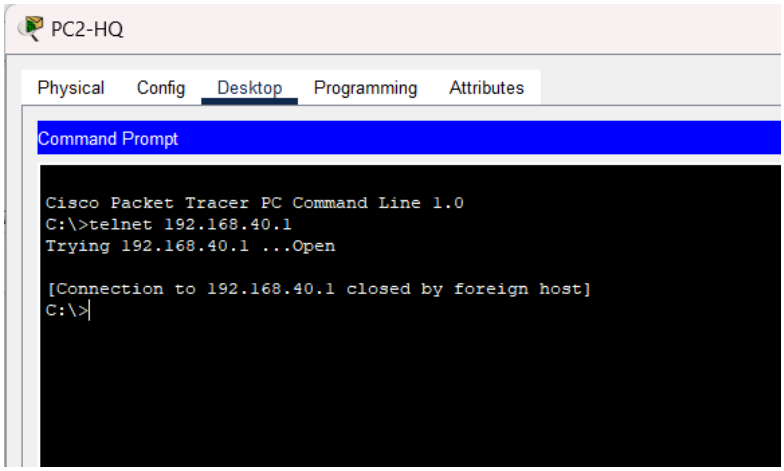


Figure 18 - Telnet DC Router from PC2 in HQ

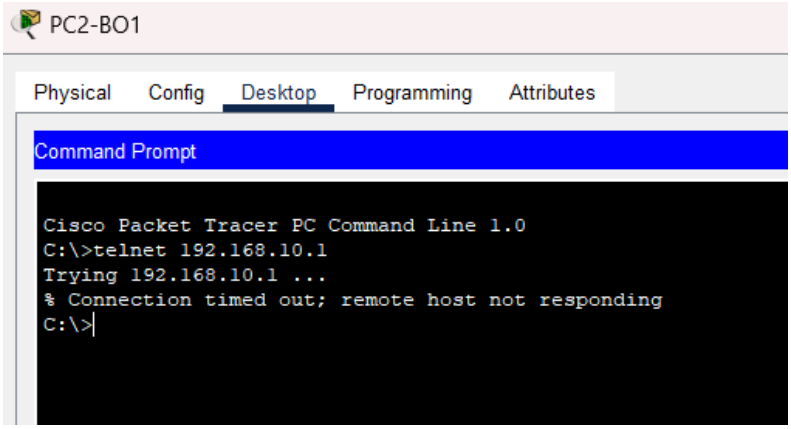


Figure 19 - Telnet Router in HQ from PC2 in BO1

Conclusion

This project effectively demonstrates the design and execution of a functional, secure, and scalable network infrastructure developed for a multi-branch enterprise. The implementation of private IP addressing, dynamic routing protocols (OSPF and BGP), and comprehensive access control measures guaranteed seamless connectivity and robust protection throughout the entire network. Every component—from routing and IP planning to firewall enforcement—was deliberately structured to improve performance, manageability, and security.

Although the current setup fulfils the organization's technical and operational objectives, potential future upgrades could involve adding advanced security layers like intrusion prevention systems (IPS), employing network segmentation via VLANs, and incorporating automation tools for streamlined configuration management. Ongoing testing, consistent updates, and security audits will be crucial for keeping the network in sync with changing business needs and new cybersecurity challenges.

References

- GeeksforGeeks (2022). *Subnetting Implementation in Cisco Packet Tracer*. [online] GeeksforGeeks. Available at: https://www.geeksforgeeks.org/subnetting-implementation-in-cisco-packet-tracer/ [Accessed 15 May 2025].
- ComputerNetworkingNotes (2018). *OSPF Configuration Step by Step Guide*. [online] ComputerNetworkingNotes. Available at: https://www.computernetworkingnotes.com/ccna-study-guide/ospf-configuration-step-by-step-guide.html [Accessed 15 May 2025].
- linuxtiwary (2018). *bgp routing configuration lab using packet tracer*. [online] Learn Linux CCNA CCNP CEH CISSP CISA Penetration-Testing Bug Bounty IPv6 Cyber-Security Network-Security Online. Available at:

https://linuxtiwary.com/2018/02/19/bgp-routing-configuration-lab-using-packet-tracer/ [Accessed 15 May 2025].

- ipcisco (2024). *3 Steps Cisco Standard ACL Configuration on Packet Tracer ⋆ IpCisco*. [online] IpCisco. Available at: https://ipcisco.com/lesson/standard-access-list-configuration-with-packet-tracer-2/ [Accessed 15 May 2025].
- Nick (2021). *Project: DMZ and Network Hardening Tutorial with Packet Tracer*. [online] Cybr. Available at: https://cybr.com/network-security-archives/project-dmz-and-network-hardening-tutorial-with-packet-tracer/ [Accessed 15 May 2025].